Predicting Late Delivery Risk in Supply-Chain Orders: A Random Forest Approach with Threshold Tuning and Permutation-Based Interpretation

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Abstract

Late deliveries damage customer satisfaction and raise logistics costs, yet many organizations learn that an order is delayed only after it has left the warehouse. This study develops a supervised model that flags orders at risk of late arrival using information that is available when the order is placed. The analysis draws on 180,519 order records, of which 54.8% were delivered late. After removing outcome-revealing, personally identifying and financial columns, three order-level features were engineered and a 200-tree Random Forest was trained inside a scikit-learn pipeline. The decision threshold was selected on a validation split carved out of the training data, and the final model was scored once on a stratified hold-out set of 36,104 orders. At a threshold of 0.40 the model achieved 75.3% accuracy, 74.7% precision, 83.1% recall, an F1 score of 78.7% and a ROC-AUC of 0.844. Permutation importance showed that payment type and order-location variables carry most of the predictive signal, while product and pricing attributes contribute very little. Dropping the scheduled shipping days changed F1 by less than half a percentage point. A three-tier categorization (Low, Medium, High) converts predicted probabilities into a simple triage tool. The paper closes with a frank discussion of limitations, including the single random split and the associational nature of importance scores.

Keywords: late delivery prediction; supply-chain analytics; Random Forest; threshold tuning; permutation importance; risk stratification

1. Introduction

Delivery reliability is one of the most visible measures of supply-chain performance. When a parcel arrives after its promised date, the buyer experiences a broken commitment, and the seller absorbs follow-up costs such as support contacts, refunds and repeat shipments. Retrospective reporting on lateness is useful for auditing, but it

cannot prevent the delay. A forward-looking alternative is to estimate, at the time an order is created, how likely it is to miss its schedule, so that planners can intervene early by re-routing, expediting or setting customer expectations differently.

This work frames that task as binary classification. Each order is labelled late (1) or not late (0), and a model learns the relationship between order attributes and that label. Three practical questions guide the study:

1. How well can late delivery be predicted from order-time attributes alone, once columns that reveal the outcome have been excluded?
2. How should the probability threshold be chosen when missing a late order and raising a false alarm carry different costs?
3. Which attributes does the fitted model actually rely on, and do the engineered features help?

The contribution is a transparent, reproducible workflow rather than a new algorithm: careful leakage control, a threshold chosen without touching the test data, a risk-tier output designed for operations teams, and an interpretation step that includes an ablation check.

2. Background

Tree ensembles. Random Forests average many decision trees grown on bootstrap samples with random feature selection at each split, which reduces variance and copes well with mixed numeric and categorical inputs (Breiman, 2001). They require little feature scaling and provide class-probability estimates through the vote share of the trees, which makes them a natural candidate for threshold-based decisions.

Target leakage. A model can look excellent in development if its inputs contain information that is only known after the outcome occurs. Kaufman, Rosset and Perlich (2012) describe this problem and its subtle forms. In delivery data, columns such as the realised shipping duration or the final delivery status are textbook examples; excluding them is a precondition for any honest evaluation.

Thresholds and imbalance of errors. Accuracy at a default cut-off of 0.50 hides the trade-off between precision and recall. Precision-oriented and recall-oriented views of performance can lead to different conclusions, and the appropriate operating point depends on the cost of each error type (Saito and Rehmsmeier, 2015).

Model interpretation. Permutation importance measures how much a performance metric deteriorates when one feature's values are randomly shuffled, evaluated on held-out data (Breiman, 2001; Fisher, Rudin and Dominici, 2019). It is model-agnostic, but it can be unreliable for correlated or high-cardinality predictors (Strobl et al., 2007), a point we return to in the discussion.

Data source. The file analysed here is named APL_Logistics.csv. Its structure (40 columns, 180,519 rows and Spanish-language place names) matches the publicly released DataCo Smart Supply Chain dataset (Constante, Silva and Pereira, 2019), which we take to be its origin.

3. Data and Methodology

3.1 Dataset and target

The raw table contains 180,519 orders described by 40 columns covering payment type, shipping mode, customer attributes, product and category fields, order geography and financial amounts. The target, Late_delivery_risk, equals 1 for 98,977 orders (54.83%) and 0 for 81,542 orders (45.17%). The classes are therefore only mildly unbalanced, and no resampling was applied.

Data quality was good. Only two columns contained missing values (Customer Lname: 8 records; Customer Zipcode: 3 records), both of which were later removed as identifying fields, and the table contained no duplicate rows. Text columns were stripped of leading and trailing whitespace.

3.2 Leakage and privacy control

Twenty-one columns were removed to leave 19 (including the target). The removed fields fall into four groups:

Group	Columns removed	Reason
Outcome-revealing	Days for shipping (real), Delivery Status, Order Status	Known only after shipment; would leak the label
Identifiers and personal data	Customer Fname, Lname, Street, Zipcode, Id; Order Customer Id; Customer City, State, Country	Privacy, and no generalisable signal

Group	Columns removed	Reason
Redundant keys and names	Category Id, Department Id, Product Name	Duplicate information already held in name columns
Financial fields	Benefit per order, Order Profit Per Order, Order Item Profit Ratio, Sales per customer, Sales, Order Item Total	Derived amounts not needed for an order-time model

The scheduled shipping duration (Days for shipment (scheduled)) was retained because it is fixed when the order is created.

3.3 Feature engineering

Three features were derived from quantity and the scheduled duration:

Feature	Definition	Distribution
Shipping_Pressure_Index	Order Item Quantity divided by (scheduled days + 1)	Numeric; most common value 0.2 (59,184 orders)
Order_Complexity	Low for 1–2 items; Medium for 3–5 items; High above 5	Low 119,449; Medium 61,070 (no order exceeded five items)
Scheduled_Duration_Category	Short for 0–2 days; Medium for 3–4 days; Long above 4	Medium 107,752; Short 72,767

No missing or infinite values were produced. After engineering, the table had 22 columns, of which 21 served as predictors.

3.4 Preprocessing and model

The 21 predictors comprised nine numeric and twelve categorical variables. Numeric columns were median-imputed. Categorical columns were imputed with the most frequent value and one-hot encoded; categories seen fewer than 50 times were grouped together and unseen categories at prediction time were ignored. These steps, together with the classifier, were wrapped in a single scikit-learn Pipeline (Pedregosa et al., 2011) so that every transformation is fitted on training data only.

The classifier was a Random Forest with 200 trees, a minimum of two samples per leaf and a fixed random seed of 42.

3.5 Evaluation protocol

The data were split 80/20 with stratification on the target, giving 144,415 training and 36,104 test orders. To choose the decision threshold without consulting the test set, the training portion was split again (80/20, stratified) into 115,532 fitting and 28,883 validation orders. A forest trained on the fitting portion produced validation probabilities, and thresholds from 0.30 to 0.60 in steps of 0.05 were compared on precision, recall and F1 for the late class. The selected threshold was then applied to a forest retrained on the full training set and evaluated once on the test set. Reported metrics are accuracy, precision, recall, F1 and ROC-AUC.

3.6 Interpretation

Feature relevance was assessed with permutation importance on the test set, using F1 as the scoring metric and three shuffles per feature. Importance is reported per original column, so one-hot columns are shuffled together. A second model without the scheduled shipping days was trained to check how much that variable contributes.

4. Results

4.1 Exploratory findings

Standard Class dominates shipping (107,752 orders, 59.7%), followed by Second Class (35,216; 19.5%), First Class (27,814; 15.4%) and Same Day (9,737; 5.4%). Each mode maps to exactly one scheduled duration: Same Day to 0 days, First Class to 1, Second Class to 2 and Standard Class to 4 days. Payment types are led by debit (69,295; 38.4%), then transfer (49,883; 27.6%), payment (41,725; 23.1%) and cash (19,616; 10.9%). Consumers account for 51.8% of orders, corporate customers 30.3% and home offices 17.9%. By market, Latin America (28.6%) and Europe (27.8%) are the largest, followed by Pacific Asia (22.9%), USCA (14.3%) and Africa (6.4%).

The historical late rate differs only modestly across the 23 order regions. It ranges from 48.8% in Canada to 58.0% in Central Africa, a spread of about 9 percentage points around the overall rate of 54.8%. Apart from Canada and Central Africa, every region lies within

three points of the overall rate. Geography alone is therefore a weak separator, which suggests that any predictive value comes from combinations of attributes rather than a single dominant factor.

4.2 Baseline at the default threshold

At the conventional cut-off of 0.50, the Random Forest scored 74.38% accuracy, 85.00% precision, 64.69% recall and 73.47% F1 on the test set, with a ROC-AUC of 0.844. The high precision and modest recall show that the default cut-off is conservative: it labels an order late only when the trees agree strongly, and therefore misses roughly one late order in three.

4.3 Threshold selection

Table 1 reports validation performance for the late class across candidate thresholds.

Threshold	Precision	Recall	F1
0.30	0.593	0.972	0.737
0.35	0.650	0.917	0.761
0.40	**0.726**	**0.821**	**0.770**
0.45	0.796	0.710	0.750
0.50	0.833	0.631	0.718
0.55	0.853	0.584	0.694
0.60	0.867	0.556	0.678

Table 1. Validation-set precision, recall and F1 by threshold.

Precision rises and recall falls steadily as the threshold increases. F1 peaks at 0.40, which was therefore adopted. At this setting the validation accuracy was 73.17%, with 12,997 late orders correctly flagged and 2,839 missed.

4.4 Final test performance

Applying the 0.40 threshold to the retrained model gave the results in Table 2.

Metric	Value
Accuracy	75.31%
Precision (late)	74.71%
Recall (late)	83.11%
F1 (late)	78.69%
ROC-AUC	0.844

Table 2. Test-set performance at threshold 0.40.

The confusion matrix contained 10,738 true negatives, 5,570 false positives, 3,343 false negatives and 16,453 true positives. The model therefore identified 83.1% of the 19,796 late orders while raising false alarms on 34.2% of the 16,308 on-time orders. Compared with the default cut-off, recall improved by 18.4 percentage points and F1 by 5.2 points, at the cost of 10.3 points of precision. Test accuracy (75.31%) was slightly higher than validation accuracy (73.17%), a difference consistent with the larger training set used for the final model and with ordinary sampling variation.

4.5 Risk tiers

To make the output usable by planners, predicted probabilities were grouped into three tiers: Low (below 0.30), Medium (0.30 to below 0.60) and High (0.60 or above). On the test set, 4,313 orders (11.9%) fell in Low, 19,052 (52.8%) in Medium and 12,739 (35.3%) in High. These cut-offs were set by judgement rather than optimised, and the paper does not claim tier-level precision because it was not computed.

4.6 Feature importance

Table 3 lists the permutation importance (decrease in F1) of the ten most influential features.

Rank	Feature	Importance
1	Type (payment type)	0.0382
2	Order City	0.0354
3	Order State	0.0322

Rank	Feature	Importance
4	Longitude	0.0237
5	Order Country	0.0212
6	Latitude	0.0211
7	Order Region	0.0150
8	Customer Segment	0.0149
9	Market	0.0122
10	Order Item Discount	0.0017

Table 3. Permutation importance on the test set.

Nine features account for nearly all of the measured signal; every remaining attribute contributes less than 0.002. Discount rate, category name and department name follow with values of roughly 0.0013 to 0.0014. Of the engineered variables, only Order_Complexity appears among the top fifteen, and its importance (0.0004) is small. Four variables showed negative importance: Days for shipment (scheduled) at −0.0183, Scheduled_Duration_Category at −0.0064, Shipping_Pressure_Index at −0.0053 and Shipping Mode at −0.0012. A negative value means that shuffling the feature slightly improved F1, which indicates that the model gains nothing from it and may be slightly harmed by its noise.

4.7 Ablation: removing scheduled shipping days

Because the scheduled duration was the most negative contributor, a second forest was trained without it. Table 4 compares the two models on the test set at the same 0.40 threshold.

Metric	With scheduled days	Without scheduled days
Accuracy	0.7531	0.7481
Precision	0.7471	0.7415
Recall	0.8311	0.8300
F1	0.7869	0.7832

Metric	With scheduled days	Without scheduled days
ROC-AUC	0.8440	0.8405

Table 4. Effect of dropping Days for shipment (scheduled).

Every metric moved by less than 0.6 percentage points, and F1 fell by only 0.37 points. The variable is therefore not a meaningful driver of the predictions. An important caveat applies: Shipping Mode, Shipping_Pressure_Index and Scheduled_Duration_Category all remained in the reduced model, and each encodes the same information, since the mode-to-days mapping is one-to-one. The ablation shows that the explicit column is redundant, not that scheduled duration is irrelevant in principle.

5. Discussion

Predictive quality. A ROC-AUC of 0.844 indicates that the model ranks late orders above on-time orders far more often than chance, yet accuracy near 75% shows that a quarter of orders remain misclassified. For an operational screening tool this is a reasonable but not exceptional result, and it is best used to prioritise attention rather than to make automatic decisions.

Why the threshold matters. The default cut-off yielded an attractive 85% precision but missed about 35% of late orders. If an unnoticed late delivery is costlier than a precautionary check, the 0.40 setting is preferable, since it recovers most late orders while keeping precision near 75%. Organisations with limited intervention capacity could instead use a higher cut-off or focus on the High tier alone.

What drives the predictions. Payment type was the strongest single feature, and geography in its several forms (city, state, country, coordinates, region, market) dominated the rest. These results describe what the model uses, not what causes delay. A payment method is unlikely to slow a parcel directly; it may act as a proxy for customer or order characteristics that the dataset records indirectly. Likewise, the weak regional late-rate differences in Section 4.1 imply that location matters mainly in combination with other fields. Practitioners should treat the ranking as a guide for further investigation, for example through targeted analysis of specific cities or payment types, rather than as proof of mechanism.

Engineered features. The three derived variables added little. This outcome is informative in its own right: the model had already access to the underlying quantity and scheduling information, and the simple ratios and bins did not expose new structure.

6. Limitations

1. **Single random split.** All figures come from one stratified split. Cross-validation or repeated splits would give confidence intervals, and a time-ordered split would test whether the model holds up on future orders.
2. **One algorithm, fixed settings.** Only a Random Forest with fixed hyperparameters was evaluated. Gradient-boosted trees or regularised linear models may perform differently, and no hyperparameter search was run beyond the threshold.
3. **High-cardinality categoricals.** Order City and Order State have many levels. One-hot encoding with a minimum frequency of 50 limits sparsity, but permutation importance can be inflated or unstable for such variables (Strobl et al., 2007). Only three shuffles were used.
4. **Correlated predictors.** Shipping Mode, scheduled days and two derived features are redundant, which explains their negative importance and means individual values should not be over-read.
5. **Coordinates.** In the first records inspected, latitude and longitude matched the customer's city rather than the order destination, so the two geographic signals may describe different places than their names suggest.
6. **Label definition.** The target is a recorded risk flag; its exact construction is not documented in the notebooks, and the analysis does not examine how it relates to the dropped delivery-status fields.
7. **Uncalibrated probabilities and judgemental tiers.** Tier boundaries were not tuned, and no calibration analysis was performed.

7. Conclusion and Future Work

A Random Forest trained on order-time attributes can flag late deliveries with a ROC-AUC of 0.844. Choosing a 0.40 threshold on a validation split raised recall to 83.1% on unseen data, at an acceptable cost in precision, and a three-tier output provides a practical way to prioritise orders. Order type and location variables dominate the model, while engineered ratios and scheduled shipping days add little, and the ablation confirms that the explicit scheduled-days column is redundant.

Future work should (a) benchmark gradient boosting and logistic regression under cross-validation, (b) use a chronological split, (c) calibrate probabilities and tune tier boundaries against real intervention costs, (d) explain individual predictions with local methods such

as SHAP, and (e) pilot the High tier with dispatch teams to measure whether early action reduces actual delays.

References

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5–32.

Constante, F., Silva, F., & Pereira, A. (2019). DataCo Smart Supply Chain for Big Data Analysis. *Mendeley Data*.

Fisher, A., Rudin, C., & Dominici, F. (2019). All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning Research, 20*(177), 1–81.

Kaufman, S., Rosset, S., & Perlich, C. (2012). Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data, 6*(4), 1–21.

Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830.

Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE, 10*(3), e0118432.

Strobl, C., Boulesteix, A.-L., Zeileis, A., & Hothorn, T. (2007). Bias in random forest variable importance measures: Illustrations, sources and a solution. *BMC Bioinformatics, 8*, 25.